



Lumen Labs

BUSINESS PROFILE

Software & technology studio, engineered with intention.

Websites. Applications. Interfaces.

COMPANY OVERVIEW / 01

A focused technology studio.

Lumen Labs is a South African technology studio building software, web applications, and technical interfaces for organisations that need more than a template. The studio's work spans frontend craft and practical technical capability — from customer-facing websites and business platforms to the operator interfaces behind industrial and IT systems.

Founded and led by Siphwe Khumalo, Lumen Labs works across software development, infrastructure, and control systems — disciplines that, in practice, rarely stay neatly separated. That range is deliberate: most technical problems touch more than one of them.

WHO WE ARE / 02

Small enough to care. Technical enough to build.

Lumen Labs sits between the limits of a template and the overhead of a large agency. Every engagement is led directly by the founder — the same person who scopes the problem, builds the system, and stays responsible for it after launch.

That structure means fewer handovers, clearer decisions, and work that remains maintainable long after delivery. Clients work with the engineer who actually understands their system, not an account layer sitting between them and the build.

01

Craft matters

Strong structure and considered details make digital work easier to trust.

02

Technical clarity

Build what is useful, keep it understandable, and avoid complexity for its own sake.

03

Direct collaboration

A focused working relationship keeps decisions clear and momentum practical.

WHAT WE DO / 03

Six disciplines, one point of contact.

Lumen Labs works across software, infrastructure, and control systems. That range is the point: most technical problems do not stay neatly inside one of them.

01

Software Development

Custom applications shaped around a real workflow instead of a generic template — the operational spine of a business, not a bolt-on tool.

02

Web & PWA Development

Installable, offline-tolerant web applications and business websites that behave like software rather than a brochure.

03

IT & Technical Support

Day-to-day technical support that looks for the underlying cause rather than clearing the same symptom on repeat.

04

Networking & Infrastructure

Wired and wireless networks, cloud hosting, and Microsoft 365 environments that stay predictable as a business grows.

05

Automation & Systems Integration

Control logic and operator interfaces for plant and facility environments, plus integration that lets existing systems exchange data without manual re-entry.

06

Business Technology Solutions

Practical hardening, CCTV and access systems, VoIP and communications, and the hardware a business depends on daily.

TECHNICAL EXPERTISE / 04

The tools, and where they sit.

Technology is a means, not a badge. This is what Lumen Labs works with and how the layers connect in practice.

LANGUAGES & FRAMEWORKS		DATA & INTEGRATION	
TypeScript	HTML & CSS	REST APIs	Webhooks
JavaScript	Node.js	JSON	Authentication
React	Python	SQL databases	Reporting pipelines
PLATFORMS & SYSTEMS		NETWORKS & CONTROL	
Windows	Cloud hosting	TCP/IP	BACnet
Linux	Virtualisation	VLANs & routing	MQTT
Microsoft 365	Backup & recovery	VPN	SNMP
		Modbus	SIP

SIGNAL PATH — REFERENCE ARCHITECTURE		
01	Field layer	Sensors · controllers · I/O
02	Transport	Modbus · BACnet · MQTT
03	Gateway	Normalise · buffer · authenticate
04	Application	REST · database · rules
05	Interface	Dashboard · alarms · history

OUR APPROACH / 05

From an idea, a system, or a problem.

Lumen Labs can start from a blank page, an existing system that needs extending, or infrastructure that is quietly costing time. The route through is the same.

01

Discover**Understand the actual problem**

Walk the process, see the current system, and find out where time is really being lost.

02

Design**Decide the shape before building**

Agree the structure, the interface, and the boundaries of the work while changes are still cheap.

03

Build**Write it properly the first time**

Small, reviewable increments with something usable to look at early rather than a reveal at the end.

04

Integrate**Connect it to what exists**

Make the new system exchange data with current equipment, accounts, and tooling.

05

Deploy**Put it into real conditions**

Release carefully, with access, backups, and a way back if something behaves unexpectedly.

06

Support**Stay responsible for it**

Monitor, maintain, and keep improving it once real use starts revealing the edges.

PROJECTS / CASE STUDIES / 06

Problems, and what was actually done about them.

Written the way an engineer would describe them: the problem, the approach, the technology, and what changed.

ARC Glasshouse

INDUSTRIAL AUTOMATION / SCADA INTERFACE

PROBLEM

Growing conditions inside a glasshouse are controlled by equipment that reports in its own language, so reading true system state meant checking several places and trusting memory for anything historical.

OUTCOME

Zone conditions and equipment state read at a glance instead of assembled from separate sources; history is available without exporting anything.

APPROACH

Map what an operator needs to decide, then design backwards from that decision — grouped by zone, with abnormal values surfaced rather than buried in a log.

Technology SCADA · Industrial controllers · Modbus · Web interface · Charting

Women's safety application

MOBILE-FIRST WEB APPLICATION

PROBLEM

In a situation where someone feels unsafe, the interface itself is the obstacle — anything requiring several taps, a login, or a steady hand is useless at the moment it matters.

OUTCOME

The critical action stays one tap from open, and the application degrades gracefully instead of failing when connectivity drops.

APPROACH

Design for the worst conditions first: one hand, low light, poor signal, no time. Keep the primary action reachable on the first screen.

Technology Progressive web app · Geolocation · Offline-first storage · Push notifications · REST API

Internal IT service desk

INTERNAL PLATFORM / TICKETING

PROBLEM

Support requests arriving by message, email, and corridor conversation could not be prioritised, and recurring faults were impossible to spot with nothing written down centrally.

OUTCOME

Requests are queued and attributable rather than remembered; recurring faults are visible in the record instead of anecdotal.

APPROACH

Give every request a single front door and a status that means the same thing to everyone, with a logging step fast enough that people actually use it.

Technology Web application · Relational database · Role-based access · Email notifications · Reporting

Ingebo Enhle Tax and Business Solutions

BUSINESS WEBSITE

PROBLEM

A credible financial services practice was difficult to verify online, putting the burden of proof on every first conversation.

OUTCOME

The practice reads as established and current, and enquiries begin with fewer basic questions about what it offers.

APPROACH

Lead with what the practice does and who it serves, in the order a prospective client actually asks, kept shallow so nothing important sits three clicks deep.

Technology Responsive frontend · Semantic HTML · Performance budget · SEO foundation

Spartcon Technologies

DIGITAL PRESENCE

PROBLEM

A technically capable business was presenting itself in a way that understated what it could actually deliver.

OUTCOME

The presentation now matches the technical level of the business rather than working against it.

APPROACH

Establish a digital presence that matches the technical standard of the work, using structure and restraint rather than volume of content.

Technology Responsive frontend · Design system · Accessibility · Metadata

Note — "SafeHold" and "Alerta" were named as reference projects for this profile but do not currently appear under those names in Lumen Labs' published project record. The case studies above are the studio's documented, sourced work; this section can be updated with SafeHold/Alerta details once confirmed.

WHY LUMEN LABS / 07

Looking good is the easy half.

A polished interface over an unreliable system is a liability. The parts that decide whether a build survives contact with real use are mostly invisible.

01 Reliability

Something that works on the day it ships is a demo. It has to still work on an ordinary Tuesday in a year.

03 Security

Access, patching, and backups treated as part of the build rather than a task for later.

05 Integration

Systems are judged by how well they cooperate with everything already in the building.

07 Real-world problem solving

The measure is whether the problem is gone, not whether the solution was interesting to build.

02 Practical engineering

The right solution is the simplest one that genuinely holds, not the most impressive one available.

04 Scalability

Room to grow where growth is likely, without paying for architecture the business will never need.

06 Maintainability

Written so the next person — possibly another developer entirely — can pick it up and continue.

CONTACT DETAILS / 08

Start a conversation.

DIRECT CONTACT

Founder	Siphiwe Khumalo
Title	Founder
Email	Lumenlabs.creatives@gmail.com
Phone	+27 63 877 6984

WEBSITE & LOCATION / 09

Website	lumenlabcreatives.spartangroup.co.za
Based	South Africa · Remote
Social	Not yet published — to be added once live

HAVE A TECHNICAL PROBLEM WORTH SOLVING?

Let's build the solution.

Start a project, discuss a problem, or simply get in touch — every route reaches the same inbox.